

Invited: Conformal Inference meets Evidential Learning: Distribution-Free Uncertainty Quantification with Epistemic and Aleatoric Separability

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ABSTRACT

This paper introduces a lightweight framework for quantifying uncertainty in deep learning models deployed at the edge, addressing the challenge of making reliable predictions under computational constraints. By integrating conformal inference and evidential learning into a novel approach called *Conformalized Evidential Quantile Regression (CEQR)*, it offers a practical solution for models to assess and communicate their confidence in predictions. The method efficiently distinguishes between aleatoric and epistemic uncertainties, ensuring statistical robustness and real-time applicability on resource-limited devices. This work paves the way for safer, more reliable AI applications in critical areas by enabling models to recognize when they *don't know*.

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1 INTRODUCTION

In the rapidly evolving landscape of artificial intelligence (AI), deploying deep learning models at the edge has emerged as a critical frontier for real-time, low-latency decision-making across various applications, from autonomous vehicles to smart healthcare systems. However, a significant challenge that persists in this domain is the quantification of uncertainty in predictions made by these models. The ability of an AI model to ascertain the extent to which it can confidently evaluate an observation and express uncertainty in its predictions has become imperative, especially in safety-critical applications where decisions based on inaccurate or overconfident predictions can have dire consequences. Although prior works have explored novel devices and circuits for this they only enable Monte-Carlo-based algorithms, hence, not amenable for real-time profiling [12, 17, 20]. This work briefly explores the nuances of embedding uncertainty quantification (UQ) mechanisms in deep learning models deployed at the edge, ultimately providing a lightweight yet

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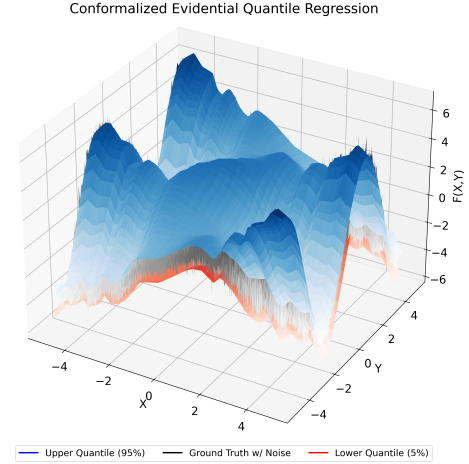


Figure 1: We propose a scalable framework leveraging conformal inference and evidential learning for robust uncertainty quantification in deep learning, utilizing quantile regression. This approach generates adaptive confidence bounds for predictions, capturing uncertainty intuitively. Consider the complex, non-linear mesh grid function $f(X, Y) = \cos(x)e^{-\frac{x^2+y^2}{20}} + x \sin(y) + \frac{\sin(x^2)}{2} + \frac{x^2+y^2}{50} + \text{noise}$ with $\text{noise} \sim \mathcal{N}(0, 2e^{-0.3\sqrt{x^2+y^2}})$ effectively encompassed by our model's conditional quantiles with an estimated marginal coverage of 92.56%.

statistically robust solution suitable for use under the unique constraints of edge devices such as limited memory, computing power, size, cost, and battery life.

Deep learning (DL), despite its impressive capabilities, is inherently plagued by its black-box nature, making the quantification of uncertainty a non-trivial task. Traditional approaches to UQ in DL, such as Bayesian Neural Networks (BNNs), offer a probabilistic interpretation of model weights; however, they are computationally intensive and often impractical for edge deployment due to the limited computational resources and power constraints of edge devices [1]. This underscores the need for lightweight, efficient, and tractable methods for edge computing.

To address these challenges, this paper proposes combining conformal inference [3, 16, 24] and evidential learning [2, 8, 15, 23], two promising, complementary methodologies for UQ in edge-deployed AI models, which is illustrated in Fig. 1. Conformal inference offers a model-agnostic, statistical framework that provides valid prediction intervals for various observations with guaranteed coverage given a finite set of samples, making it a robust tool for expressing uncertainty in predictive modeling. Evidential learning, on

the other hand, provides a sample-free framework for DL models to directly learn the parameters of an unknown target distribution through a hierarchical Bayesian structure without repetitive sampling, which can then be used to extract explicit uncertainty estimates. Combined, the methodologies offer a pathway to implementing lightweight and efficient UQ mechanisms suitable for edge intelligence applications.

2 BACKGROUND

Exploring uncertainty in DL requires overcoming significant hurdles and considerations pivotal for the practical deployment of machine learning technologies, especially within the context of edge computing. These models, while advanced, can often present considerable uncertainty in their outputs, a fact that may not be apparent to users. This poses potential risks in critical areas such as healthcare, robotics, remote monitoring, and others where reliability is paramount. The ability to measure uncertainty in these models is essential to gauge confidence in their predictions accurately.

Two primary forms of uncertainty are identified in DL: *aleatoric* and *epistemic* [9]. Aleatoric uncertainty, or statistical uncertainty, stems from inherent randomness in data. It is typically present and cannot be reduced, even in ideal modeling conditions, due to issues such as sensor errors or corrupted data. Approaches for modeling this type of uncertainty include the use of probabilistic neural networks, which provide outputs in the form of distributions and allow assessment of the variability in the data. On the contrary, epistemic uncertainty (uncertainty related to the model's knowledge), diminishes as more data becomes available. As such, it is particularly acute in scenarios where the model encounters little to no data. BNNs are commonly applied to assess epistemic uncertainty in DL by incorporating priors over weights and repeatedly facilitating posterior inference [5, 13]. Although approximations and hardware accelerators for BNNs have been explored [7, 18, 19], excessive inferential computations render them impractical for the edge.

In essence, the intricacies of uncertainty in deep learning demand a thorough and methodical strategy for quantification and mitigation. To this end, we introduce *Conformalized Evidential Quantile Regression (CEQR)*, an uncertainty quantification method that combines conformal inference with deep evidential learning. This synergy of theory capitalizes on the statistical guarantees of conformal inference in covering complex, heteroskedastic data that can be arbitrarily shaped, and the proficiency of evidential learning in assessing and modulating model confidence, especially for out-of-distribution (OOD) observations. As is discussed in Section 3, the combined technique imposes little overhead and can be adapted to almost any data-driven deep-learning regression task.

3 CONFORMALIZED EVIDENTIAL QUANTILE REGRESSION

In this section, we explore the combined uncertainty quantification methodology proposed in this work, namely Conformalized Evidential Quantile Regression (CEQR). We will outline the foundational theories of evidential learning and conformal inference, providing a detailed overview of each along with their associated loss functions. CEQR is practically implemented using a deep neural network (DNN) designed with two output heads: one for making

predictions, using Mean-Squared Error, and another for producing quantiles related to each parameter of the learned evidential distribution, facilitating the extraction of predictive uncertainty.

3.1 Evidential Learning

Evidential learning (EL) is a novel approach for separately quantifying aleatoric and epistemic uncertainties with deterministic neural networks, offering a comprehensive framework for applications involving classification, regression, *etc.* [23]. This method frames learning as a process of accumulating evidence, placing evidential priors over the likelihood function as opposed to network weights, thus streamlining the estimation of uncertainty without relying on more elaborate Bayesian methods with redundant sampling [2, 11, 15]. Its resilience in accurately recognizing and adjusting to out-of-distribution (OOD) samples further underscores its adaptability and effectiveness in maintaining performance where other models typically falter.

Our work integrates the deep evidential quantile regression method proposed in [8], which extends the original deep evidential regression method from [2], for jointly learning the target and the uncertainty by incorporating Bayesian quantile regression. The inclusion of Bayesian quantile regression allows for better accommodation of non-Gaussian distributions, broadening the applicability of this method in complex scenarios and effectively removing the need for assuming Gaussianity while estimating aleatoric uncertainty. Additionally, paired with conformal inference, this EL method transfers well into practical implementation.

To learn the parameters of the evidential prior distribution that maximizes the likelihood of observing targets Y and thus allows assessment of aleatoric and epistemic uncertainty, the regression task is framed in a hierarchical Bayesian structure. The assumed Gaussian target distribution $\mathcal{N}$ which contains Y is parameterized with (μ, σ^2) . The parameterization is that of a scalar mixture of Gaussians based on quantiles following an asymmetric Laplace distribution [8, 10, 25] whose parameters maximize the likelihood of Y by minimizing the negative log-likelihood (NLL) function [6]. However, since μ and σ^2 are unknown, their distributions must instead be determined which involves placing a normal prior on μ , and an Inverse-Gamma prior on σ^2 . These distributions combine to form the Normal-Inverse-Gamma (NIG) evidential prior distribution parameterized with learnable parameters $G = \{\gamma, v, \alpha, \beta\}$ where generally γ represents the expected sample mean, v represents the precision, α represents the shape, and β represents the scale of the distribution [2, 4, 8].

Total evidence (i.e., model confidence) is defined as $\Phi = 2v + \alpha + \frac{1}{\beta}$ and is used as a penalty when the network makes poor predictions. This evidence is scaled with tilted loss to regularize learning the evidential distribution, balancing uncertainty inflation with model fitting. Thus, learning G involves training a neural network to maximize evidence with each proper observation and minimize it when predictions are wrong. This requires two loss functions:

$$\begin{aligned} \mathcal{L}_{NLL} = & \frac{1}{2} \log\left(\frac{\pi}{v}\right) - \alpha \log(\Omega) \\ & + \left(\alpha + \frac{1}{2}\right) \log((y - \gamma + \tau z)^2 v + \Omega) + \log\left(\frac{\Gamma(\alpha)}{\Gamma\left(\alpha + \frac{1}{2}\right)}\right) \end{aligned} \quad (1)$$

and

$$\mathcal{L}_{reg} = \Phi[\mathbb{I}\{y \geq \gamma\}q(y - \gamma) + \mathbb{I}\{y < \gamma\}(1 - q)(\gamma - y)], \quad (2)$$

where y is an element of all true values Y , $\Omega = 4\beta(1 + \tau z v)$, $\tau = \frac{1-2q}{q(1-q)}$, $z \sim \text{Exp}\left(\frac{1}{\beta/(\alpha-1)}\right)$, and q is the desired quantile level. A balancing hyperparameter λ_{reg} is applied to $\mathcal{L}_{reg}$ and the two losses are summed.

We can then express aleatoric and epistemic uncertainty along with predictions as below:

$$\underbrace{\mathbb{E}[\mu] = \gamma}_{\text{prediction } \hat{y}}; \quad \underbrace{\mathbb{E}[\sigma^2] = \frac{\beta}{\alpha-1}}_{\text{aleatoric uncertainty}}; \quad \underbrace{\text{Var}[\mu] = \frac{\beta}{v(\alpha-1)}}_{\text{epistemic uncertainty}}. \quad (3)$$

3.2 Conformal Inference

Utilizing quantile regression within the framework of evidential learning, as outlined in the preceding discussion, provides us with an algorithm that is inherently uncertainty-aware, making it a compelling choice for enhancing uncertainty quantification techniques. Nonetheless, quantile regression by itself, despite its adaptability to data with varying heteroskedasticity, does not ensure satisfactory marginal coverage for a target variable y within the corresponding posterior distribution Y . To address this limitation, the algorithm can be paired with conformal inference to theoretically improve its statistical efficiency. Conformal inference (CI), a lightweight statistical methodology, offers reliable, distribution-free confidence intervals for predictive outcomes, regardless of the data's underlying distribution. It employs historical data to assess the conformity of new data points within a predictive framework.

By integrating CI with EL, we employ the CI-based quantile regression approach proposed by [22], termed conformalized joint prediction (CJP). More extensions of our CI-based quantile regression for multi-mode and multi-band quantification can be found in [14, 21]. This strategy, given input variables X , concurrently trains a model to produce multivariate point estimates as well as lower and upper conditional quantile bounds. These bounds are highly adaptive and can be shaped freely, creating an uncertainty-informed confidence interval $C(X)$ that aims for a 90% marginal coverage of Y as its objective. This process is facilitated by a calibration dataset, which, in traditional conformal inference methods, would be separated from the training dataset and utilized subsequently. However, CJP proposes a novel approach, suggesting the entire training dataset could serve as a calibration tool, provided that coverage is estimated and averaged across random training batches. This integration is particularly beneficial for EL, ultimately enhancing quantile precision while offering protection against out-of-distribution (OOD) instances.

Instead of estimating a single value for each parameter of the learned evidential distribution, we estimate and conformalize the 5^{th} and 95^{th} quantiles of each. This allows for explicit interpretations of aleatoric and epistemic uncertainty as $[\gamma^{95^{th}} - \gamma^{5^{th}}]$ and $[|2\sigma^{95^{th}}| + |2\sigma^{5^{th}}|]$ respectively where σ is the standard deviation of the evidential mean and is obtained by square root of $\text{Var}[\mu]$.

Conformalization of the quantiles is achieved with the following loss function:

$$\mathcal{L}_{cal} = (1 - \lambda_{cal}) \times \text{COV}_{obj} + \lambda_{cal} \times \text{SHARP}_{obj} \quad (4a)$$

where

$$\text{COV}_{obj} = \mathbb{I}\{p_{avg}^{cov} < p\}[(y - \hat{q})\mathbb{I}\{y > \hat{q}\} + \mathbb{I}\{p_{avg}^{cov} > p\}[(\hat{q} - y)\mathbb{I}\{y < \hat{q}\}], \quad (4b)$$

$$\text{SHARP}_{obj} = \mathbb{I}\{p \leq 0.5\}[(1 - \hat{q}) - \hat{q}] + \mathbb{I}\{p > 0.5\}[\hat{q} - (1 - \hat{q})], \quad (4c)$$

$\hat{q}$ is the predicted conditional quantile, p_{avg}^{cov} represents the average probability that each target y in a training batch falls under each $\hat{q}$, and λ_{cal} is a hyperparameter balancing coverage of Y (COV_{obj}) with minimizing the distance between $\hat{q}$ and $1 - \hat{q}$ in covering Y (SHARP_{obj}). This function ensures that the network calibrates the 5^{th} and 95^{th} quantiles associated with the learned evidential distribution towards $p = 90\%$ marginal coverage of targets Y over each training batch, establishing the confidence interval $C(X)$ around each prediction. Additionally, we seek to empirically center Y within the computed $C(X)$ interval to enhance its accuracy. To do this, we add loss on q to follow the desired quantile level:

$$\mathcal{L}_{interval} = (\hat{q}^{95^{th}} - \hat{q}^{5^{th}}) + \mathbb{I}\{y < \hat{q}^{5^{th}}\} \frac{2}{q} (\hat{q}^{5^{th}} - y) + \mathbb{I}\{y > \hat{q}^{95^{th}}\} \frac{2}{q} (y - \hat{q}^{95^{th}}). \quad (5)$$

4 RESULTS

The uncertainty quantification framework proposed in Section 3 is evaluated here. Fig. 2 demonstrates separate quantification of aleatoric and epistemic uncertainties associated with predicting the meshgrid function from Fig. 1. This 500×500 non-linear meshgrid function was chosen specifically to be a relatively complex test for a simple DNN. It includes polynomial, trigonometric, and exponential terms, requiring precise prediction of 250,000 points with similarly complex additive noise.

As is shown, the function is extended from the training range of $[-5, 5]$ to $[-15, 15]$ in the x - y plane to demonstrate increased uncertainty (i.e., reduced confidence) upon processing unfamiliar samples in the purple region. In Fig. 2a, we observe that the aleatoric uncertainty has a small peak in the $[-5, 5]$ range due to the additive noise being most present around the origin and exponentially decaying elsewhere. It increases considerably beyond this original range (which is indicated by the grey rectangle), translating to a widened gap between the lower 5^{th} and upper 95^{th} conformalized evidential sample mean quantiles. In Fig. 2b, we observe a similar result as the variance in each quantile (which are combined in the visualization) sharply increases beyond $[-5, 5]$.

To summarize, the CEQR algorithm involves learning quantiles of the parameters of an evidential prior distribution with which we can separately and explicitly quantify aleatoric and epistemic uncertainty. The quantiles are then conformalized (i.e., calibrated) so as to guarantee marginal coverage of and optimally adapt to the ground truth. Of course, coverage decreases with new OOD observations, however, through evidential learning, the DNN model can clearly indicate that it does not know whether its prediction is correct regardless of how well it performs. These benefits are

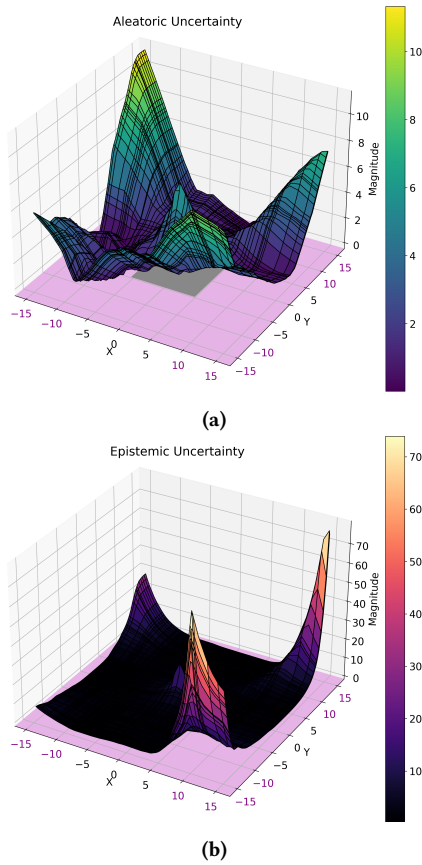


Figure 2: For the function in Fig. 1, we provide separable (a) aleatoric and (b) epistemic uncertainty meshgrids based on a fresh test set. The function was extended from $[-5, 5]$ to $[-15, 15]$ in the x-y plane to demonstrate heightened uncertainty (i.e., decreased confidence) upon OOD samples. The purple x-y plane shading depicts this extended range whereas the grey rectangle covers the original function range.

not restricted to specific tasks on platforms with ample computing resources; the framework is scalable and widely applicable, making it suitable for edge applications.

5 CONCLUSION

We presented a novel lightweight, scalable approach that combines conformal inference and evidential learning to enable precise quantification of aleatoric and epistemic uncertainties and effectively communicate confidence in predictions. The framework's ability to offer statistically robust, real-time applicable uncertainty assessments makes it a pivotal tool for enhancing the reliability and safety of AI applications in resource-constrained environments. **Code will be released at:** <https://github.com/acstutts/CEQR.git>

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